



Eaton

Eaton 93Li G2 Lithium Battery System



Powering Business Worldwide

Eaton 93Li Lithium Battery System

Eaton always dedicates to providing customers with high-quality power supply products and services, as well as an optimized total power solution. The solution realizes the perfect integration of UPS system, high power lithium battery system with intelligent battery monitoring system. The one-step solution highlights a strict control over the production process and testing and adopts Eaton-tailored BMS protocol to better guarantee the reliability of customers' power supply systems. The selected high power lithium battery system features excellent safety performance, high energy density, long lifecycle, good thermal management performance, and environmental- friendly design. It perfectly matches power system demand from high-end manufacturing, large data centers, utility, telecom, and other application segments.

Product Features

- **High safety:** LiFePO₄
- **High reliability:** 3 level BMS system (cell-level,module-level,system-level)
- **High rate:** 6C discharge
- **Intelligent communication:** multi-level protection of battery system through communication integration with UPS
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- **Scalability on-site:** support 8 racks in parallel, faster deployment and configuration on-site
- **Modular design:** easy replacement
- **Pre-assembled shipment:** reducing transportation and construction costs
- **Design calendar life 15 years at 25°C environment condition**
- **Compact cabinet design to save footprint**
- **Environmentally-friendly**

Applications

- High-end manufacturing
- Large data centers
- Utility
- Mobile power station

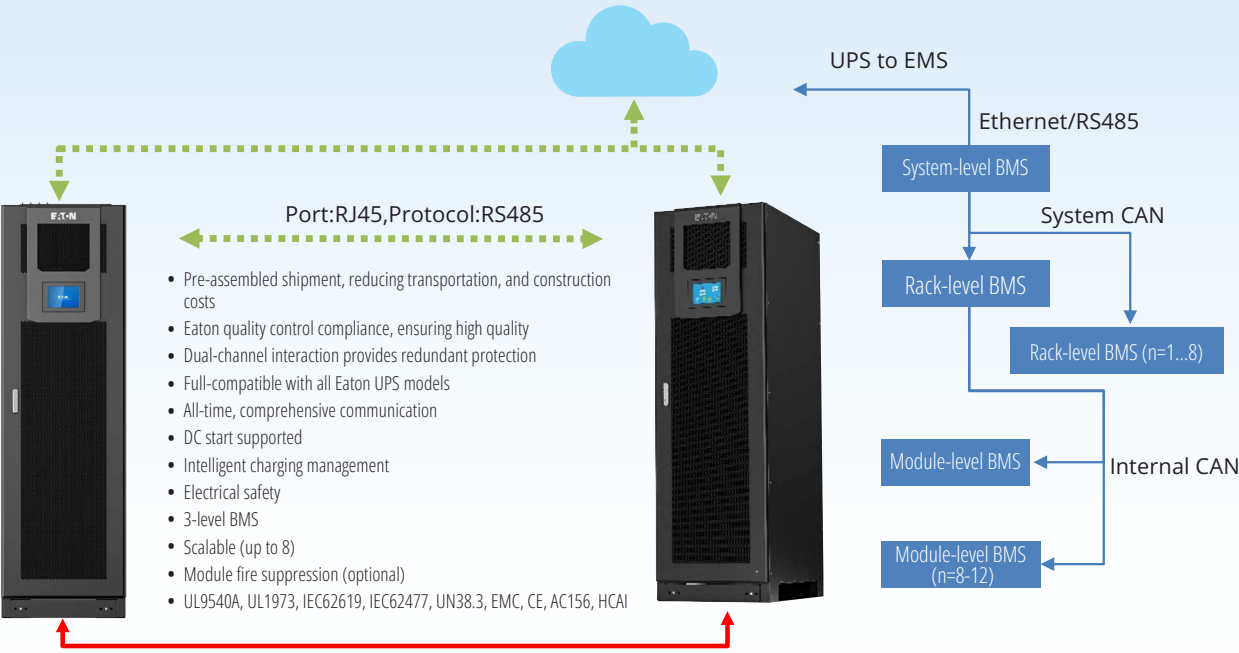
BMS Functions

- Charging and discharging current protection
- Software and hardware protection
- Independent control of charging and discharging circuit
- Internal equalization
- Reverse voltage protection
- Monitoring and protection at cell-level, module-level, and system-level
- SOC/SOH available
- Multiple Communication (CAN/RS485/dry contact)
- Control and protection in parallel operation

Communication

Communication interface (RS485/CAN) provides battery operating status and fault information.

Eaton UPS and Lithium Battery Solution



Cell & Module Technical Specification

	Cell	Module
Rated voltage (V)	3.2	51.2
Rated Capacity (Ah)	52	104
Rated Energy (kWh)	0.166	5.324
Series & Parallels	-	16S2P
Dimensions (W*D*H)	122*13.0*360	442*752*129
Weight (kg)	1.08	54

Rack Technical Specification

	8-Module/RACK	9-Module/RACK	10-Module/RACK	11-Module/RACK	12-Module/RACK	8-Module/RACK	10-Module/RACK	12-Module/RACK
	93Li128S-100Ah-2PBFA	93Li144S-100Ah-2PBFA	93Li160S-100Ah-2PBFA	93Li176S-100Ah-2PBFA	93Li192S-100Ah-2PBFA	93Li128S-100Ah-3PBFA	93Li160S-100Ah-3PBFA	93Li192S-100Ah-3PBFA
Series & Parallels	128S2P	144S2P	160S2P	176S2P	192S2P	128S2P	160S2P	192S2P
Rated Voltage (V)	409.6	460.8	512	563.2	614.4	± 204.8	± 256	± 307.2
Rated Capacity (Ah)	104							
Rated Energy (kWh)	42.598	47.923	53.248	58.572	63.897	42.598	53.248	63.897
Charge voltage (V)	441.6	496.8	552	607.2	662.4	± 220.8	± 276	± 331.2
Charging current (recommended/maximum)	0.25C / 1.0C*							
Discharge voltage (V)	358.4	403.2	448	492.8	537.6	± 179.2	± 224	± 268.8
Max. Discharge Power (kW)	215.04	241.92	268.8	295.68	322.56	215.04	268.8	322.56
Wiring	2 wire					3 wire		
Dimensions (W*D*H)	600*900*2000							
Weight (kg)	755	810	865	915	970	755	865	970
Communication	CAN (Racks Parallel)/ RJ45(between UPS and Racks)							
Altitude	≤3000m (above 2000m,derating)							
Storage temperature/Humidity	-30℃~-20℃ & 45℃~60℃ for 7 days, -20℃~ 45℃ for six months, Related humidity:5%-95%RH							
Operating temperature	0~40℃ (Recommended operating temperature:20~30℃)							
Compatible UPS	Eaton 9395P 200~1200kVA, Eaton 9395XR 1.5~1.8MW, 91PS 8-30, 93PS 8-40, Eaton 93PR 25-200kW, 93PR 200~1200kW, Eaton 9EHD industrial machine Third-party 2-wire UPS with certain configuration*					Eaton 93T 15~200kVA Third-party 3-wire UPS with certain configuration*		
Certificates	UL9540A, UL1973, IEC62619, IEC62477, UN38.3, EMC, CE, AC156, HCAI							

Note: 1、 Only dry contact communication is supported for third-party UPS. 2、 0.25C (0°C≤T≤15°C; 45°C≤T≤60°C); 1.0 C (15°C≤T≤45°C) 3、 Contact with Eaton to get detailed product configuration.

